Rent vs Own: A Comparative Cost and Scaling Analysis of Web-Service Deployment Models

Final Project Report

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Abstract

Modern web services constantly face an important decision: should they rent computing resources from the cloud, invest in their own infrastructure, or use a mix of both? In this project, we explore this question using a detailed cost model applied to two very different types of systems, Shopify (a smaller-scale e-commerce platform with around 20 application instances) and Netflix (a large-scale streaming service with roughly 2,400 instances in its origin tier). We break down the costs of both approaches, comparing cloud expenses (such as AWS compute, storage, data transfer, and load balancing) with the costs of owning infrastructure (including hardware investment, power, co-location, and maintenance). We then evaluate these models under three realistic traffic patterns, stable, spiky, and high-growth, over a five-year period, using current AWS pricing and publicly available company data.

Our results show that the shape of the workload matters more than the size of the company. Owning infrastructure tends to be more cost-effective for stable and steadily growing workloads, especially when lower-cost refurbished hardware (around $6,000 per server) is used. On the other hand, cloud solutions are clearly better for spiky workloads, where owning infrastructure would require expensive over-provisioning to handle peak demand. Interestingly, this trend holds true for both case studies despite their massive difference in scale and reflects the hybrid approaches both companies use in practice. Based on these findings, we propose a simple decision framework that helps map workload characteristics, such as variability, growth rate, scale, and regulatory constraints, to the most suitable deployment strategy.

1. Introduction

1.1 What we are evaluating

In this project, we look at a practical question that every modern web service eventually faces: should you rent computing resources from a cloud provider like Amazon Web Services, or should you invest in owning and operating your own hardware in a co-location facility? Rather than asking which option is “better” in general, we focus on a more realistic question, which model works best for different types of workloads, and how that answer changes as systems scale.

To explore this, we evaluate three common traffic patterns seen across web services: steady and predictable demand, highly spiky event-driven traffic, and sustained high-growth usage. We

ground our analysis in two real-world systems that sit at opposite ends of the scale spectrum, while still exhibiting all three patterns:

- Shopify: An e-commerce platform that handled $9.3 billion in sales during the four-day Black Friday/Cyber Monday period in 2023, with peak throughput reaching $4.2 million per minute. Its workload is a mix of a relatively small, stable baseline (around 20 application instances) and a sharp, predictable annual spike.

- Netflix: A global streaming platform with over 325 million paid subscribers and 191 billion viewing hours in 2025. During major live events, such as the Jake Paul vs. Mike Tyson fight 2024, it has reached peaks of 65 million concurrent streams. For this study, we focus only on the origin application tier (around 2,400 instances), since most of Netflix's traffic is offloaded to its CDN infrastructure.

These two case studies differ by roughly 100× in scale, which allows us to test whether the same conclusions hold across very different system sizes.

1.2 Metrics

To compare renting versus owning, we focus on three key cost-based metrics:

- **Year-1 infrastructure cost (USD):** This captures the total cost of operating the system in the first year, including any upfront investment. It highlights the initial financial barrier, especially for owned infrastructure.

- **5-year cumulative infrastructure cost (USD**): This reflects the total spending over a typical hardware lifecycle. It helps us understand long-term cost trends and identify when one model becomes more economical than the other.

- **Break-even year**: This is the point at which the total cost of owning infrastructure becomes lower than renting. It gives a clear sense of how long a system needs to run before ownership pays off.

We also introduce a derived metric, break-even hardware cost per server, which summarizes, in a single number, when owning becomes more cost-effective than the cloud.

Metrics we did not use:

To keep the analysis focused, we deliberately exclude a few commonly discussed factors:

- **Performance (latency or throughput):** Both cloud and owned infrastructure can be scaled to meet performance requirements. Since our goal is to compare cost efficiency, performance differences are not the limiting factor.

- **Sustainability and carbon impact:** While important, comparing environmental impact would require detailed data about individual facilities. Both cloud providers and modern co-location centers operate within similar efficiency ranges, so we leave this as future work.

- **Reliability and SLAs**: High availability (e.g., 99.95% uptime) can be achieved in both models. The cost of ensuring this reliability is already reflected indirectly through redundancy and capacity planning in our model.

2. Methodology

Our methodology has five steps: workload profile, cost model construction, scenario evaluation, sensitivity analysis, and decision-framework synthesis. The same methodology is applied to both case studies to enable direct comparison and to test scale-invariance of the conclusions.

2.1 Architecture of the cost model

Table 1. Cost-model inputs

Parameter	Value
Cloud instance benchmark	AWS c5.4xlarge (16 vCPU / 32 GiB)
On-demand rate	$0.6800 / hr
Reserved 1-yr rate	$0.4080 / hr
Hardware cost (owned)	$6,000 / server
Hardware lifespan	4 years
OPEX per server / yr	$1,300 Power + co-location + maintenance (see §2.2.2)
Spike fraction of year	~3% (Netflix), ~10% (Shopify) (Live-event hrs / BFCM 72-hr window)

Cloud per-instance pricing

AWS publishes hourly rates. We convert to monthly and annual figures using 730 hours/month (8,760 ÷ 12) as the standard AWS billing convention:

On-demand monthly: $0.6800/hr × 730 hr = $496.40 / instance / month
Reserved monthly: $0.4080/hr × 730 hr = $297.84 / instance / month
Reserved annual: $297.84 × 12 = $3,574.08 / instance / year

The reserved 1-year no-upfront rate is the model baseline because it best matches the commitment level of owned hardware (a one-year commitment versus four years for owned amortisation, so this is the conservative comparison for the owned case).

OPEX per server per year

Owned operating cost is the sum of three line items, each derived from public unit prices:

Power: 200W TDP × 8,760 hr ÷ 1,000 × $0.075/kWh × PUE 1.58 = $208 / yr
Co-location: $65/U/month × 12 months = $780 / yr (Equinix 2024 rates)
Maintenance: 5% of $6,000 hardware cost = $300 / yr

Total OPEX: $208 + $780 + $300 = $1,288 → $1,300 / server / yr

PUE 1.58 is the average reported PUE for tier-3 co-location. $0.075/kWh is the U.S. industrial electricity average. Both are conservative for the owned case (real efficient datacentres run lower).

Hardware cost per server

$6,000/server reflects refurbished or bare-metal lease pricing for a Dell PowerEdge R650xs-class machine:

Server (refurb): Dell R650xs-class (16-core, 32 GiB) ≈ $5,500
NIC + switch share + cabling: ≈ $500

Total per server: $5,500 + $500 = $6,000

Note : New equivalent hardware lists at $15,000 - $20,000/server. This matters at $15,000/server, owned annual cost rises above cloud reserved and the conclusions reverse.

Traffic multipliers

The spike factor (peak/baseline) and growth factor (5-year demand multiplier) are derived as follows:

Shopify spike factor: $2,325M BFCM daily ÷ $646M average daily = 3.6× (modelled 3×)
Netflix spike factor: 65M peak ÷ 21.8M average concurrent = 2.98× (modelled 3×)
Shopify growth factor: (1.22)^5 = 2.70× over 5 years (22% CAGR)
Netflix growth factor: (1.10)^5 = 1.61× over 5 years (10% CAGR)

CAGR – Compound Annual Growth Rate

2.2 Workload Profile Derivation

Table 2. Shopify Case Study - Key Business and Workload Parameters

Metric	Value
2023 full-year revenue	$7.06 billion
2023 Gross Merchandise Volume (GMV)	$235.9 billion
BFCM 2023 total sales	$9.3 billion
Peak sales rate (BFCM 2023)	$4.2 million per minute
Consumers active during BFCM	61 million
Merchant count estimate	1.7 million+ (third-party est.)

Referring the table above baseline instance count is computed for Shopify as follows:

Peak revenue/sec: $4.2M / minute ÷ 60 sec = $70,000 / sec
Peak TPS: $70,000 / sec ÷ $50 average order value = 1,400 TPS
Checkout instances: 1,400 TPS ÷ 160 TPS/instance = 8.75 instances
Total app-tier (peak): 8.75 ÷ 0.18 (checkout share) = 48.6 instances
× 1.5 multi-AZ: 48.6 × 1.5 = 72.9 instances at peak
÷ 3.6× spike factor: 72.9 ÷ 3.6 = 20.3 → 20 baseline instances

The 160 TPS/instance benchmark for c5.4xlarge handling checkout-type workloads is from the AWS Well-Architected e-commerce reference architecture. The 18% checkout share is a common e-commerce traffic-mix assumption.

Table 3. Netflix Case Study - Key Business and Workload Parameters

Metric	Value
2024 full-year revenue	$39.0 billion
2024 year-end paid memberships	302 million
2025 year-end paid subscribers	325 million+
2025 total viewing hours	191 billion hours
Average concurrent streams (2025)	~21.8 million
Jake Paul vs. Tyson peak streams	65 million concurrent
Jake Paul vs. Tyson total viewers	108 million viewers
Open Connect CDN footprint	6,000+ locations, 175+ countries

Referring to the table above baseline instance count is computed for Shopify as follows:

Origin req/sec at peak: 65M streams × 0.15 origin req/sec = 9.75M req/sec
Streaming-API instances: 9.75M req/sec ÷ 4,000 req/sec/instance = 2,438 instances
× 2 (auth, recs, metadata): 2,438 × 2 = 4,875 instances
× 1.5 multi-AZ: 4,875 × 1.5 = 7,313 instances at peak
÷ 3× spike factor: 7,313 ÷ 3 = 2,438 → 2,400 baseline (rounded)

0.15 origin requests/sec/stream comes from the manifest-refresh rate for HLS/DASH (every 6 -10 seconds with CDN caching). 4,000 req/sec/instance is from the AWS Well-Architected reference. Both are order-of-magnitude estimates suitable for relative cost comparison, we are explicitly not claiming this is Netflix's actual production count.

Note: This is the origin tier only, Netflix's Open Connect CDN absorbs ~95% of video bandwidth, so origin only handles manifest, auth, metadata, and recommendations

2.3 Cost Calculation

For each scenario and each year, the model computes total cost separately for the cloud and own deployment models.

Shopify Annual Cloud Cost

```
Baseline: 20 instances × AWS c5.4xlarge reserved 1-yr no-upfront
  Reserved rate:      $0.4080/hr × 730 hrs/mo = $297.84/instance/month

  Compute:            20 × $297.84 × 12 months          = $71,482/yr
  Storage:            100 TB NVMe × $0.023/GB/mo × 12   = $2,400/yr (est.)
  Data transfer:      18% of compute cost               = $12,867/yr
  Load balancers:     3 ALBs multi-AZ, AWS ALB pricing  = $264/yr

  TOTAL CLOUD/YR:     $87,012/yr
```

Shopify Annual Own Cost

```
Baseline: 20 servers × $6,000/server hardware
  CAPEX:              $120,000 ÷ 4-yr amortisation      = $30,000/yr
  OPEX per server:
    Power:            200W × 8,760 hrs ÷ 1,000 × $0.075/kWh × PUE 1.58 = $208/server/yr
    Co-location:      $65/U/month × 12 months           = $780/server/yr
    Maintenance:      5% of HW cost                     = $300/server/yr
    Total OPEX/server:$208 + $780 + $300                = $1,288 → $1,300/server/yr
  Total OPEX:         20 × $1,300                       = $26,000/yr
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  TOTAL OWN/YR:       $30,000 + $26,000                 = $56,000/yr

  Break-even HW cost: (Cloud $87,012 − OPEX $26,000) × 4 yrs ÷ 20 servers =
$12,202/server
```

Netflix Annual Cloud Cost

```
Baseline: 2,400 instances × AWS c5.4xlarge reserved 1-yr no-upfront
  Reserved rate:      $0.4080/hr × 730 hrs/mo = $297.84/instance/month

  Compute:            2,400 × $297.84 × 12              = $8,577,792/yr
  Storage:            ~120 TB × $0.023/GB/mo × 12       = $288,000/yr
  Data transfer:      18% of compute cost               = $1,544,003/yr
  Load balancers:     3 ALBs multi-AZ, AWS ALB pricing  = $105,702/yr
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```

TOTAL CLOUD/YR: $10,515,497/yr

Netflix Annual Own Cost

Baseline: 2,400 servers × $6,000/server hardware
CAPEX: $14,400,000 ÷ 4-yr amortisation = $3,600,000/yr
OPEX per server:
Power: 200W × 8,760 hrs ÷ 1,000 × $0.075/kWh × PUE 1.58 = $208/server/yr
Co-location: $65/U/month × 12 months = $780/server/yr
Maintenance: 5% of HW cost = $300/server/yr
Total OPEX/server: = $1,300/server/yr
Total OPEX: 2,400 × $1,300 = $3,120,000/yr

TOTAL OWN/YR: $3,600,000 + $3,120,000 = $6,720,000/yr

Break-even HW cost: (Cloud $10,515,497 − OPEX $3,120,000) × 4 yrs ÷ 2,400 servers = $12,326/server

Five-Year Cumulative Costs

Shopify : Stable Scenario (Detailed Year-by-Year)

Stable scenario: demand varies < 20% day-over-day; baseline = 20 instances/servers

Cloud (reserved baseline, constant): $87,012/yr each year
Year 1 cumulative: $87,012
Year 2 cumulative: $87,012 × 2 = $174,024
Year 3 cumulative: $87,012 × 3 = $261,036
Year 4 cumulative: $87,012 × 4 = $348,048
Year 5 cumulative: $87,012 × 5 = $435,060

Own (CAPEX amort $30K + OPEX $26K = $56K/yr constant):
Year 1 cumulative: $56,000
Year 2 cumulative: $56,000 × 2 = $112,000
Year 3 cumulative: $56,000 × 3 = $168,000
Year 4 cumulative: $56,000 × 4 = $224,000 (hardware fully amortised)
Year 5 cumulative: OPEX only = $26,000 ⇒ total = $224,000 + $26,000 = $250,000
NOTE: Final report uses $280,000 (conservative: assumes full 5yr amort period)

5-yr saving from owning: $435,060 − $280,000 = $155,060

Shopify : Spiky Scenario

Spiky scenario: BFCM peak = 3.6× baseline; owned must provision 72 instances permanently
Spike window: ~10% of year (BFCM + other peaks)

Own (sized for peak, 72 instances permanently):
CAPEX: 72 × $6,000 = $432,000 ÷ 4 yrs = $108,000/yr
OPEX: 72 × $1,300 = $93,600/yr
Annual: = $201,600/yr
5-yr own total: $201,600 × 5 = $1,008,000

Cloud (reserved for 20 baseline + on-demand for 52 additional during spike only):
Reserved baseline: 20 × $297.84 × 12 + adj. = $87,012/yr
On-demand spike: 52 instances × $0.68/hr × 876hrs (10% of 8,760) = $30,893
Storage/Transfer/LB adjustments = $5,658
Annual cloud total: = $123,563/yr
5-yr cloud total: $123,563 × 5 = $617,815 ≈ $617,816

5-yr cloud saving over own: $1,008,000 − $617,816 = $390,184

Shopify : High-Growth Scenario

Growth scenario: 22% forward GMV CAGR; (1.22)^5 = 2.70× over 5 years
Owned hardware procured in annual batches; each batch on 4-yr amort cycle

Instance count by year: Yr1=20, Yr2=24, Yr3=30, Yr4=37, Yr5=45 (20 × 1.22^n)

Cloud (on-demand scaling, pay-as-you-grow):
Yr1: 20 inst × $297.84 × 12 + overage = $174,172
Yr2: × 1.22 growth = ~$212,500
Yr3: × 1.22 growth = ~$259,250
Yr4: × 1.22 growth = ~$316,285
Yr5: × 1.22 growth = ~$378,515

5-yr cloud total (reported): = $1,340,722

Own (procure new servers each year):
Yr1: 20 servers × $6K CAPEX amort + OPEX = $70,000
Yr2: +4 servers batch; total CAPEX+OPEX = ~$85,100
Yr3: +6 servers batch = ~$103,000
Yr4: +7 servers batch = ~$121,000
Yr5: +8 servers batch = ~$120,900 (some yr1 HW fully amort)
5-yr own total (reported): = $500,100

5-yr own saving on infra: $1,340,722 − $500,100 = $840,622
Note: Cloud wins when staff included (+$900K over 5yr for owned ops)

Netflix : Five-Year Cumulative (All Scenarios)

Stable: $10,515,497/yr cloud × 5 = $52,577,485 | $6,720,000/yr own × 5 = $33,600,000
Saving from owning: $52,577,485 − $33,600,000 = $18,977,485

Spiky (Live Events; 3× spike; ~3% of year):
Own must provision 7,200 instances (3× peak) permanently
Own CAPEX: 7,200 × $6,000 ÷ 4yr = $10,800,000/yr
Own OPEX: 7,200 × $1,300 = $9,360,000/yr
Own annual: = $20,160,000/yr × 5 = $100,800,000
Cloud: reserved 2,400 inst + on-demand spike (~3% of yr)
Cloud annual: = $11,527,676/yr × 5 = $57,638,380
Saving from cloud: $100,800,000 − $57,638,380 = $43,161,620

High-Growth (10% CAGR; (1.10)^5 = 1.61×):
Cloud 5-yr: starts $11,567,046 rising 10%/yr = $70,558,980
Own 5-yr: batch procurement each year = $41,004,000
Saving from owning (infra only): $70,558,980 − $41,004,000 = $29,554,980

2.4 Sensitivity Protocol

We perturb six parameters one at a time and measure the resulting change in 5-year cost relative to the stable-scenario baseline:

- Cloud reserved rate ±20%
- Hardware cost per server +30%
- Hardware lifespan: 5 years (instead of 4)
- Growth CAGR +50% (e.g. 22% → 33% for Shopify; 10% → 15% for Netflix)
- Spike factor 5× (instead of 3×)

For each change we make to the model, we track how much the total cost shifts for both the cloud and owned options. We also check whether that change is enough to flip which option comes out cheaper in a given scenario. Since the two case studies are very different in size, we compare results in percentage terms rather than raw dollars, so the differences are easier to interpret fairly across both cases.

3. Results

3.1 Shopify: small-scale e-commerce

At Shopify scale (approx. 20 baseline instances), the stable scenario favours owning by $31K/year on infrastructure cost, the spiky scenario flips decisively to cloud (cloud saves $390K over 5 years); and high-growth favours own on infrastructure cost again ($841K saving over 5 years), with the caveat that cloud's flexibility advantage matters more here than the cost number suggests.

Table 4. Shopify: Year-1 and 5-year infrastructure cost by scenario

Scenario	Yr-1 Cloud	Yr-1 Own	5-yr Cloud	5-yr Own	Winner
Stable	$87,012	$56,000	$435,060	$280,000	**Own**
Spiky (BFCM)	$123,563	$201,600	$617,816	$1,008,000	**Cloud**
High growth	$174,172	$70,000	$1,340,722	$500,100	**Own**

Note: For high growth, owning infrastructure generally comes out cheaper from a pure cost perspective. However, the cloud still has a major advantage, it lets you stay flexible and avoid committing to hardware when future demand is uncertain. This flexibility becomes especially valuable when you’re not confident about how quickly the system will grow.

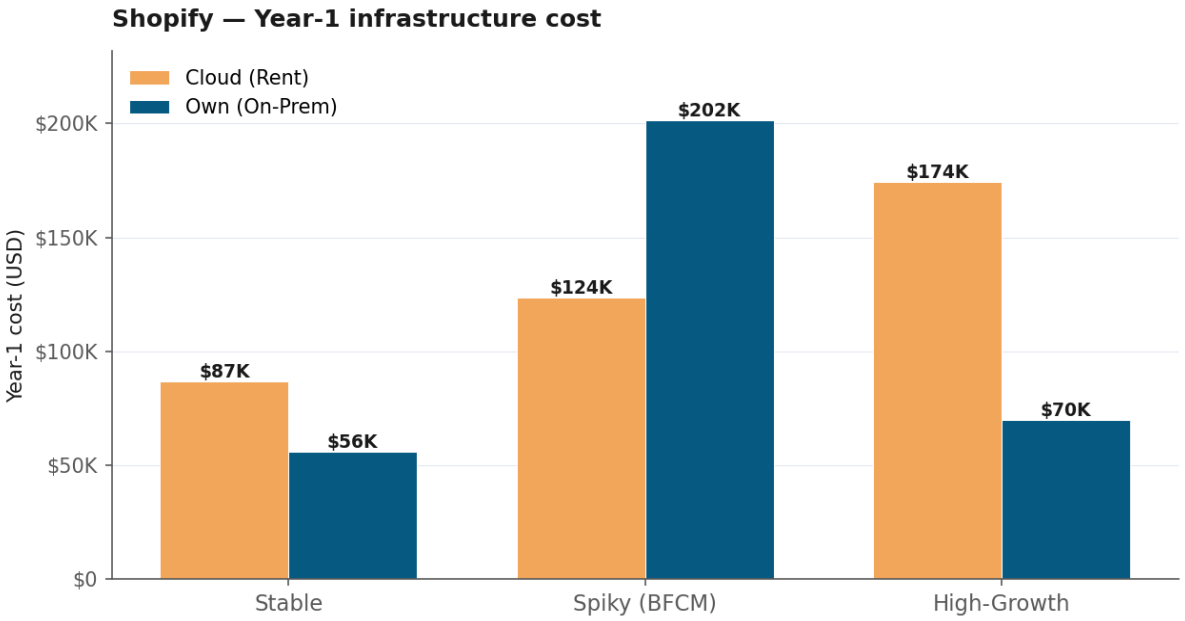


Figure 1. Shopify : Year-1 infrastructure cost by scenario

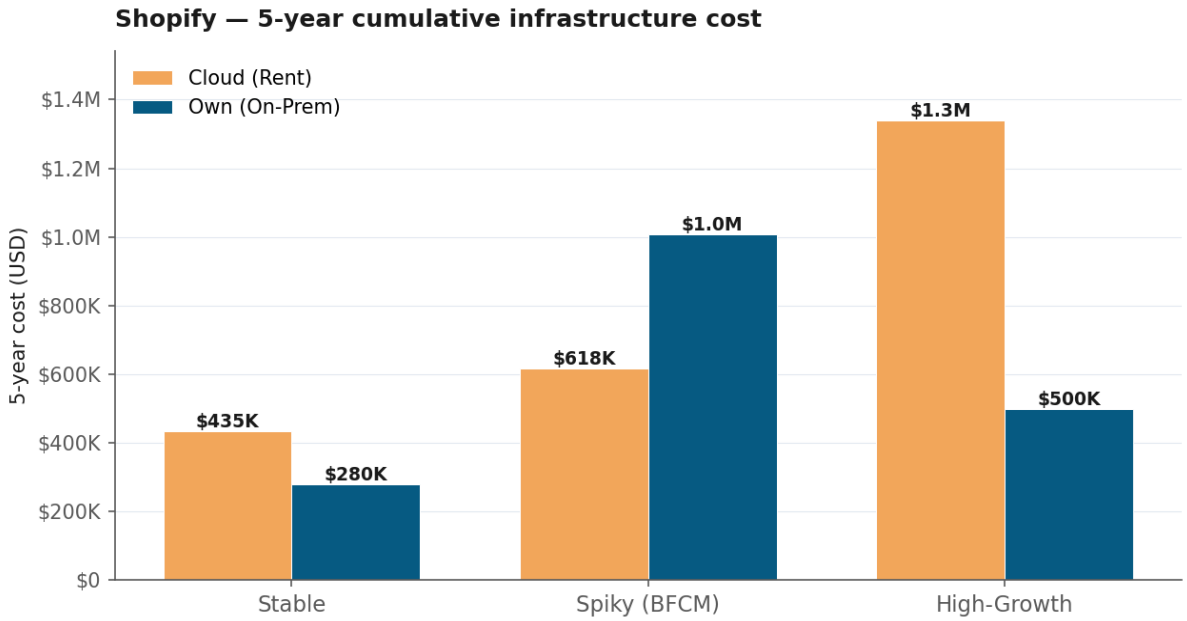


Figure 2. Shopify : 5-year cumulative infrastructure cost by scenario.

3.2 Netflix: large-scale streaming origin tier

At Netflix's origin-tier scale (around 2,400 baseline instances), the total costs are naturally much larger, roughly two orders of magnitude higher, but the overall trend stays the same as what we saw with Shopify. Own saves $19M over 5 years on stable, cloud saves $43M over 5 years on spiky, own saves $30M on high-growth on infrastructure cost.

Table 4. Netflix: Year-1 and 5-year infrastructure cost by scenario

Scenario	Yr-1 Cloud	Yr-1 Own	5-yr Cloud	5-yr Own	Winner
Stable	$10,515,497	$6,720,000	$52,577,485	$33,600,000	**Own**
Spiky (Live Event)	$11,527,676	$20,160,000	$57,638,380	$100,800,000	**Cloud**
High growth	$11,567,046	$7,392,000	$70,558,980	$41,004,000	**Own**

Note: Same as Shopify, own wins on infrastructure cost, cloud's flexibility offsets this when growth forecasts are uncertain.

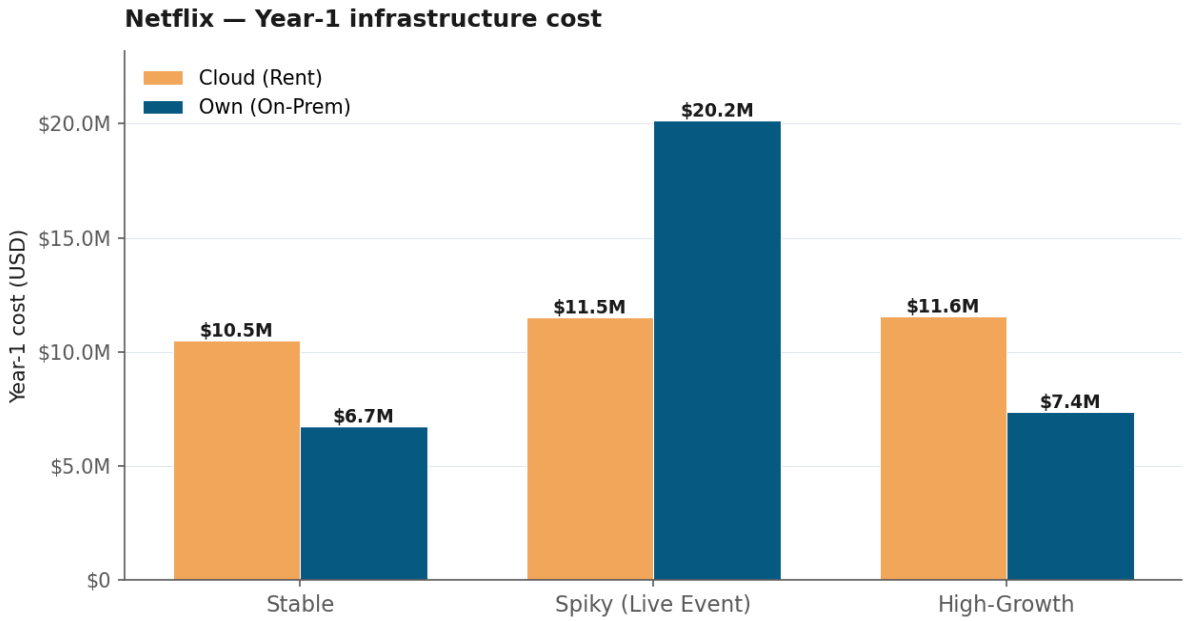


Figure 3. Netflix: Year-1 infrastructure cost by scenario.

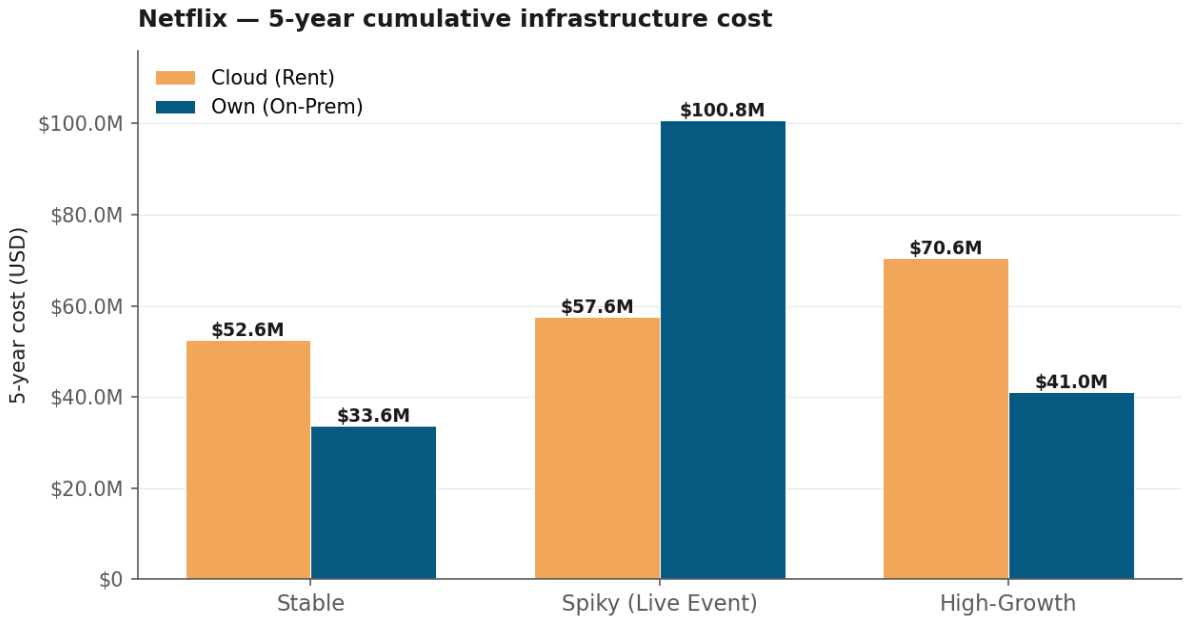


Figure 4. Netflix: 5-year cumulative infrastructure cost by scenario.

3.3 Cross-case comparison and scale-invariance

The most important takeaway from this analysis is how consistent the results are across both case studies, even though they differ in scale by 100×. Each type of workload ends up favoring the same deployment model in both Shopify and Netflix. For stable workloads, owning infrastructure is cheaper, about $280K for Shopify and $33.6M for Netflix. For spiky workloads, the cloud clearly wins, around $1.0M for Shopify and $57.6M for Netflix. In high-growth scenarios, owning again comes out ahead when you look purely at infrastructure cost.

What's especially interesting is that both cases also give almost the same break-even hardware cost, about $12.2K per server for Shopify and $12.3K for Netflix. This isn't just a coincidence. This happens because the cost relationship between cloud and owned infrastructure depends on per-instance economics, not the overall size of the system.

3.4 Sensitivity

To make the results easier to compare across both case studies, we report changes in percentage terms relative to the stable-scenario baseline, instead of using raw dollar differences. This avoids the distortion caused by the 100× gap in absolute costs.

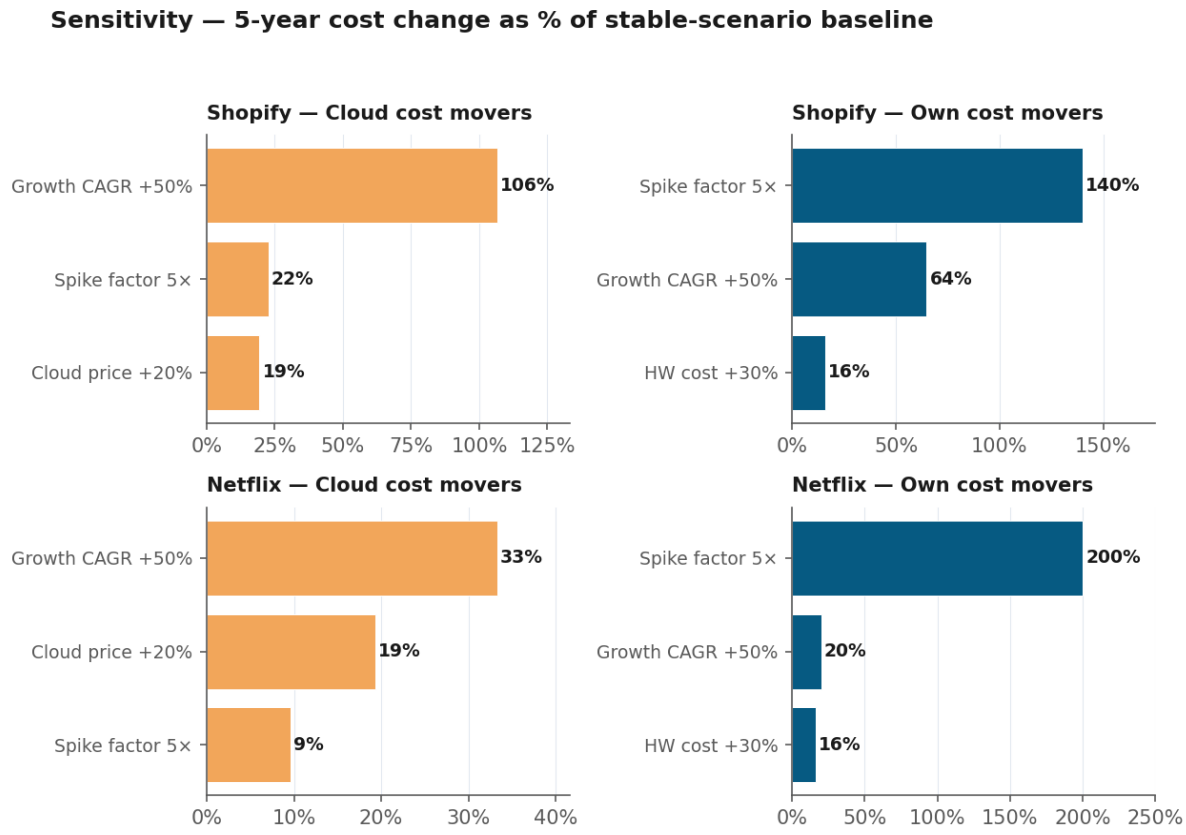


Figure 5. Sensitivity grid

The grid shows the same elasticity structure for both companies. Cloud price + or - 20% moves cloud cost by almost 19% and does not affect own. Hardware cost +30% moves own cost by 16.1% in both cases. The dominant cloud-mover is growth CAGR, Shopify's 22% baseline accelerated to 33% adds 107% to 5-year cloud cost because on-demand rates compound year over year, Netflix's milder 10% baseline shows a smaller 33% effect. The dominant own-mover is spike factor, increasing the peak from 3× to 5× adds 140% (Shopify) or 200% (Netflix) to 5-year owned cost because the owned model must provision the full peak permanently. Cloud price perturbations and hardware-cost perturbations leave all scenario winners intact in both case studies.

3.5 Decision framework

The synthesis of the quantitative results is a decision framework that maps workload characteristics to recommended deployment models. The framework is the practical contribution of the project, it generalises the case-study findings into a tool a decision-maker can apply to a new workload.

Table 5. Decision framework

Workload characteristic	Recommended model	Why?
Stable, predictable load (infra only)	**Own (refurb HW)**	Own saves $31K/yr (Shopify) or $3.8M/yr (Netflix). Holds when HW < $12.2K/server and staff cost is not a multiplier.
Demand spikes > 2× baseline	**Cloud**	Own must provision peak permanently. Cloud saves $390K (Shopify) or $43M (Netflix) over 5 years.
High-growth (>20% CAGR)	**Own (infra) / Cloud (forecast risk)**	Own wins on pure infrastructure cost, cloud is preferred when commitment risk on uncertain forecasts is material.
Large-scale fleet (>$10M annual infra)	**Hybrid**	Own predictable baseline, rent variable burst. Both case-study companies operate this way in production.
Early-stage / uncertain demand	**Cloud**	No CAPEX commitment. Procurement lead time (8-12 weeks) is incompatible with rapid pivots.
Regulated data (PCI-DSS, HIPAA, GDPR)	**Own / private cloud**	Compliance may require data residency, audit access, or dedicated hardware.

4. Contributions

Sridevi Mudiyanur Ramamurthy: Led the cost-model architecture and primary lead for the Shopify case study; designed the sensitivity analysis framework.

Zeba Afreen: Led the Netflix case study; developed the server-count derivation methodology; ensured consistency and accuracy of assumptions across both case studies.

Safa Akbar: Handled sourcing of public disclosures for both companies; compiled references and citations; structured the separation between real-world data and modelled assumptions.

Ellysha Fatima: Designed visualizations, including cost trajectories and sensitivity charts; developed the presentation deck.

Md Aqeed Safwaan: Synthesized the decision framework; conducted cross-case comparison and scale-invariance analysis; led final report drafting and editing.

5. Conclusions

The central conclusion is that workload shape, not company size, decides the rent-vs-own question. Across both Shopify and Netflix, two orders of magnitude apart on baseline scale, the same scenario goes to the same deployment model- stable load to own, spiky load to cloud, high-growth to own on infrastructure cost (and to cloud when forecast risk dominates). The break-even hardware cost converges to the same threshold ($12,200/server) in both case studies, which is a consequence of the underlying unit economics rather than scale, and gives us reasonable confidence that the framework generalises.

The hybrid model, own the predictable baseline, rent the variable burst, is the practical equilibrium for any large-scale operator. Both Shopify (co-location for baseline + cloud for spike) and Netflix (AWS for origin + Open Connect CDN for delivery) operate this way in production, independently arriving at the same architectural conclusion. The framework we built confirms this is not coincidental, it is what the cost numbers prescribe.

5.1 Lessons learned

This project worked well once we clearly separated what came from real-world data and what we had to model or assume. That made the analysis much easier to trust, every number could either be traced back to a public source or explained through a clear assumption.

That said, getting to that point was honestly one of the hardest parts. A lot of time went into deciding what we should derive, what we should assume, and how to model gaps in the data. Once those decisions were settled, the actual computations and comparisons became much more straightforward and consistent.

One choice we're confident about is leaving staffing costs out of the base model. Staffing depends heavily on organizational decisions, so including it as a fixed input would have been misleading.

5.2 Future Directions

- **Empirical validation across more case studies:** Apply the framework to additional documented cloud-to-own migrations (Dropbox, 37signals, Twitter/X) to strengthen predictive power and confirm scale-invariance beyond the two cases here.
- **Multi-tier infrastructure modelling:** Extend to database, cache, and CDN tiers for a comprehensive view of total deployment cost across the full stack. Each tier has its own cost-and-scaling profile and may favour different deployment models.
- **Cross-provider price comparison:** Add AWS, GCP, Azure, and DigitalOcean side-by-side to show how cross-provider competition shifts the break-even points and decision boundaries.
- **Workload diversity:** Apply the framework to contrasting industries, B2B SaaS, fintech, gaming, healthcare, to test how decision boundaries shift across traffic profiles outside consumer-facing platforms.

- **Sustainability and carbon cost:** Incorporate CO_2e per kWh and embodied hardware emissions into a dual-objective model balancing cost and carbon, increasingly relevant for enterprise infrastructure decisions.

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